

Name: _____

TRANSFORMATIONS, (REFLECTION)

Date: _____

LO: I can reflect shapes in mirror lines including the axes and the lines $y = x$ and $y = -x$.

Reflection

A **reflection** flips a shape across a **mirror line**. Each point and its image are the same distance from the mirror line.

Common mirror lines: x-axis ($y = 0$), y-axis ($x = 0$), $y = x$ (diagonal), $y = -x$, and vertical/horizontal lines like $x = 2$.

Quick examples

●	Reflect (3, 2) in y-axis gives (-3, 2)	x changes sign
●	Reflect (3, 2) in x-axis gives (3, -2)	y changes sign
●	Reflect (3, 2) in $y = x$ gives (2, 3)	swap x and y
●	Reflect (3, 2) in y-axis gives (3, -2)	that is reflection in x-axis

Key idea

Each point is the same perpendicular distance from the mirror line as its image, on the opposite side.

(a, b) in y-axis (-a, b)

Reflecting in the y-axis negates the x-coordinate

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - reflection in y-axis

Reflect (4, 3) in the y-axis.

Mirror line is $x = 0$ (y-axis)

x-coordinate changes sign: 4

becomes -4

y-coordinate stays the same: 3

(-4, 3)

Reflection in y-axis: (x, y) becomes (-x, y).

Example 2 - reflection in x-axis

Reflect (-2, 5) in the x-axis.

Mirror line is $y = 0$ (x-axis)

x-coordinate stays: -2

y-coordinate changes sign: 5

becomes -5

(-2, -5)

Reflection in x-axis: (x, y) becomes (x, -y).

Example 3 - reflection in $y = x$

Reflect (1, 4) in the line $y = x$.

The line $y = x$ is the diagonal

Swap x and y coordinates

(1, 4) becomes (4, 1)

(4, 1)

Reflection in $y = x$ swaps the coordinates.

Example 4 - reflection in $x = 2$

Reflect (5, 3) in the line $x = 2$.

Distance from $x = 2$: $5 - 2 = 3$

Image is 3 to the left of $x = 2$

Image $x = 2 - 3 = -1$; y stays at 3

(-1, 3)

Find distance to mirror line, go same distance on other side.

YOUR TURN – PRACTICE (deliberate practice)

1. Reflection in the y-axis

Reflect each point in the y-axis.

- a) (3, 1)
- b) (-2, 4)
- c) (5, -2)
- d) (0, 7)

2. Reflection in the x-axis

Reflect each point in the x-axis.

- a) (2, 5)
- b) (-3, -1)
- c) (4, 0)
- d) (-1, 6)

3. Reflection in $y = x$ and $y = -x$

Reflect in the given line.

- a) (2, 5) in $y = x$
- b) (3, -1) in $y = x$
- c) (4, 2) in $y = -x$
- d) (-1, 3) in $y = -x$

4. Reflection in other lines

Reflect in the given mirror line.

- a) (5, 2) in $x = 3$
- b) (1, 4) in $x = 4$
- c) (2, 6) in $y = 4$
- d) (3, 1) in $y = 3$

5. Mixed reflections

Perform each reflection.

a) (4, -1) in y-axis

b) (-3, 2) in x-axis

c) (6, 1) in $x = 4$

d) (2, 7) in $y = 5$

e) (5, 3) in $y = x$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Reflecting (3, 2) in the y-axis gives (-3, 2) ☐

Correct answer: _____

Reason: _____

b) Reflecting (3, 2) in the y-axis gives (3, -2) ☐

Correct answer: _____

Reason: _____

c) Reflection in $y = x$ swaps the x and y coordinates ☐

Correct answer: _____

Reason: _____

d) Reflection changes the size of the shape ☐

Correct answer: _____

Reason: _____

e) A point on the mirror line maps to itself ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Triangle PQR has vertices P(1, 2), Q(4, 2), R(4, 5). Reflect it in the line $y = x$ to get P'Q'R'. Then reflect P'Q'R' in the **x-axis**. Describe the single transformation that maps PQR to the final image.

Expression: _____

Simplified: _____

b) Point A(5, 3) is reflected in an unknown vertical mirror line to give image A'(1, 3). Find the equation of the mirror line. Point B(2, 7) is reflected in the same line. Find B'.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

